

Contact

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www.linkedin.com/in/sambit-baliarsingh-8322bbb1 (LinkedIn)
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Top Skills

Engineering Leadership
Agentic AI Development
Data Engineering

Languages

English (Full Professional)
Hindi (Native or Bilingual)
Oriya (Native or Bilingual)

Certifications

Internet Of Things (IOT)
Python Data Structures From
University of Michigan
Advanced Agentic AI – Production
Engineering
Walmart Global Tech GitHub Copilot
Champion

Honors-Awards

Six Sigma Green Belt
Individual Excellence Award
Customer Award
VPD Award
Spot Award

Patents

A System and Business model
for initiating strategic product
improvement programs in aerospace
industries

sambit baliarsingh

AI Platforms & Applied AI | Agentic AI & LLM Systems | ML Platforms
| Distributed Systems
Bengaluru, Karnataka, India

Summary

I work at the intersection of Agentic AI, LLM systems, ML platforms, and large-scale data infrastructure, with experience designing production capabilities across RAG, evaluation, observability, security, workflow orchestration, and AI-assisted engineering. My background spans petabyte-scale data platforms, real-time streaming, distributed computing, ML/NLP systems, and enterprise AI, with hands-on experience across Spark, Kafka, Airflow, cloud platforms, LangGraph, MCP, vector search, and production ML systems. At Staff/Principal level, I focus on technical direction, architecture, engineering quality, mentoring, cross-team execution, and turning emerging AI capabilities into reliable production systems.

Experience

Walmart Global Tech India

Staff Engineer

February 2025 - Present (1 year 8 months)

Led the architecture and delivery of enterprise Agentic AI platforms for governance, metadata intelligence, lineage, and engineering automation across Walmart-scale, petabyte-class data ecosystems.

Defined and implemented production-grade LLM platform capabilities spanning RAG, LangGraph, MCP, OAuth, fine-tuning, evaluation, vector search, and observability for secure enterprise AI adoption.

Drove AI-powered Spark reliability, optimization, and upgrade initiatives for mission-critical workloads operating on petabyte-scale data and high-volume distributed platforms.

Provided technical leadership across Walmart-scale AI/ML and data platforms, shaping architecture and roadmap, mentoring engineers, and driving execution across large-scale batch and streaming systems.

Twilio

Principal Engineer

November 2022 - January 2025 (2 years 3 months)

Bengaluru

Led architecture and engineering for production ML and data systems supporting messaging intelligence, compliance, fraud/risk detection, and growth use cases.

Built and productionized NLP classification systems using transformer models and traditional ML techniques, covering model evaluation, inference, latency optimization, and production monitoring.

Designed large-scale streaming and batch ML pipelines using Spark, Kafka, EMR Serverless, Iceberg, and Airflow/MWAA for high-volume production workloads.

Improved performance, reliability, and operational cost of ML/data workloads through architecture redesign, pipeline optimization, and production engineering.

Walmart Global Tech India

Senior Engineer

July 2019 - November 2022 (3 years 5 months)

India

Architected and scaled petabyte-scale data and ML platforms supporting Sam's Club Advertising and Sponsored Products across real-time analytics, decisioning, batch, and streaming workloads.

Built production ML data pipelines covering feature engineering, model training workflows, inference integration, evaluation, and monitoring across high-volume advertising datasets.

Designed distributed systems using Spark, Kafka, Airflow, and cloud-native technologies to support large-scale data processing and machine-learning workloads.

Improved platform reliability and performance through Spark optimization, resilient orchestration, scalable architecture, and cross-functional technical leadership.

BT

Consultant/Senior Professional, S&IT

March 2016 - July 2019 (3 years 5 months)

Ipswich, United Kingdom, Bangalore

Modernized large-scale telecom data platforms using Cloudera, Spark, Scala, Python, and data lake architectures, improving scalability, performance, and analytical readiness.

Led Hadoop-as-a-Service migration initiatives, transitioning legacy RDBMS workloads to distributed big-data platforms through reusable ingestion and processing frameworks.

Built BT's NPS scoring and analytics platform, developing scalable data pipelines and scoring workflows to support customer-experience measurement, reporting, and decision-making.

Honeywell Technology Solutions, Inc.

Senior Engineer

October 2012 - March 2016 (3 years 6 months)

Greater Bengaluru Area

Built and optimized large-scale data integration and ETL/ELT pipelines across distributed systems, improving throughput, scalability, and data reliability.

Developed Hive and Python-based data processing workflows to transform data from diverse source systems while maintaining quality and consistency.

Contributed across the full SDLC, from requirements and technical design through development, testing, production support, and performance tuning.

Tecnotree Corporation

Software Engineer

June 2012 - October 2012 (5 months)

Development of data solutions using oracle ,linux.

Designed different data ingestion pipelines.

ZTE Corporation

Project Engineer

June 2009 - March 2011 (1 year 10 months)

Mumbai Metropolitan Region

Designed and developed enterprise ETL solutions, translating business requirements into scalable data processing workflows using SQL and relational data platforms.

Created functional and technical designs covering data flows, transformations, integration patterns, and implementation requirements.

Built Unix shell-based automation for ingestion, validation, transformation, and batch processing workflows.

Education

Indian Institute of Management Visakhapatnam

Executive Master of Business Administration , Leadership with AI, Artificial Intelligence · (May 2026 - March 2027)

Birla Institute of Technology and Science, Pilani
Master of Technology - MTech, Data Science and Machine Learning · (2021 - 2023)

BIJU PATNAIK UNIVERSITY OF TECHNOLOGY
Bachelor of Technology (BTech), ELECTRONICS AND TELECOMMUNICATION · (2004 - 2008)